

Session 10

Development and the Environment

Sustainability, Environmental Limits, and Degrowth

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“ *However, let us not flatter ourselves too much with our human victories over nature. For every such victory, nature takes revenge on us.*”

In the first place, each victory has the consequences we have counted on, but in the second and third place it has quite different, unforeseen effects, which only too often cancel out those first consequences.”

Friedrich Engels, Dialektik der Natur, own translation.





Source: Ney Murtha | Banco de Imagens ANA

“ On this river, so rich in magnificent natural stones, the explorer will find pleasure for a long time [...]. The animal kingdom, the plant kingdom and even inanimate nature are above the influence of the European and will retain their originality; their wealth will never run dry, and would even if Brazil's foundations were ransacked for gold and precious stone.”



Prince Maximilian of Wied-Neuwied, 1815,
z.n. Lessenich (2016): Neben uns die Sintflut

“ *It is irrational to hope, against the evidence, that our existing growth-based economic system will deliver the development outcomes we want while at the same time reversing ecological breakdown.*”



Jason Hickel (2020)



Max Roser (2021)

“ *The reason that such large economic growth is necessary to reduce global poverty is that the world is still very poor ... those who don't see the importance of growth are not aware of the extent of global poverty.*”

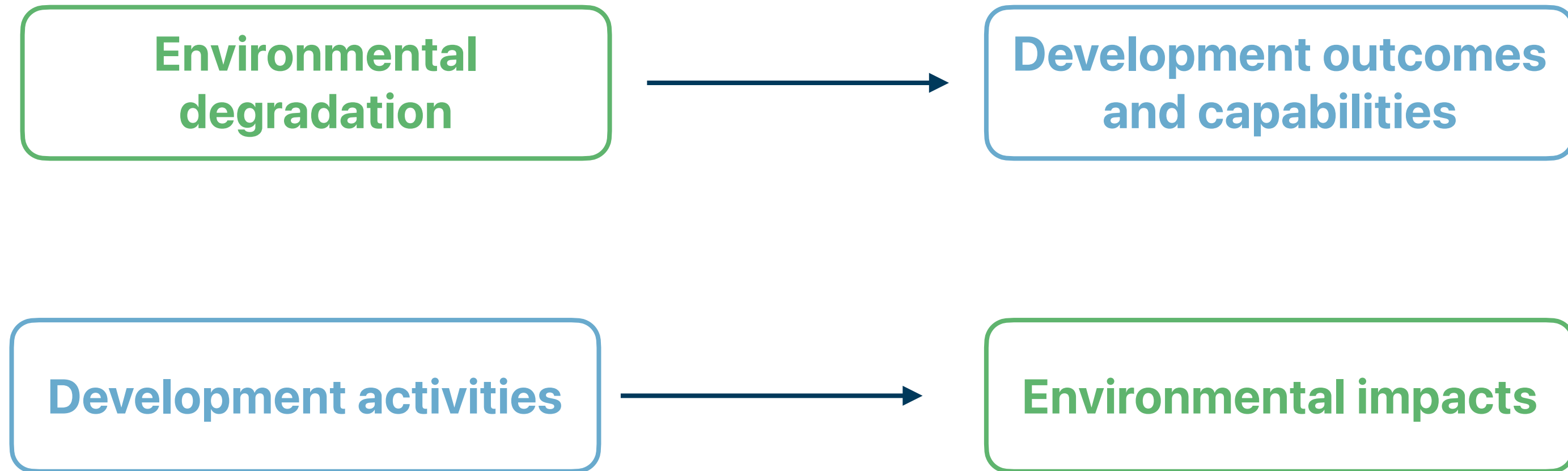
Development and the Environment

1. The Development-Environment Nexus
2. Environmentalism of the Poor
3. What the data does (not) tell us
4. The Doughnut Economics Framework
5. Growth, Green Growth, and Degrowth
6. Discussion, synthesis, and outlook

01

The Development-Environment Nexus

A bidirectional relationship



Key insight

The relationship is **not one-directional**, and is contingent on **development level**, **geography**, and **historical context**.

Three channels

Scale effect

More production → more resource use and pollution

Composition effect

Structural change → different environmental footprint

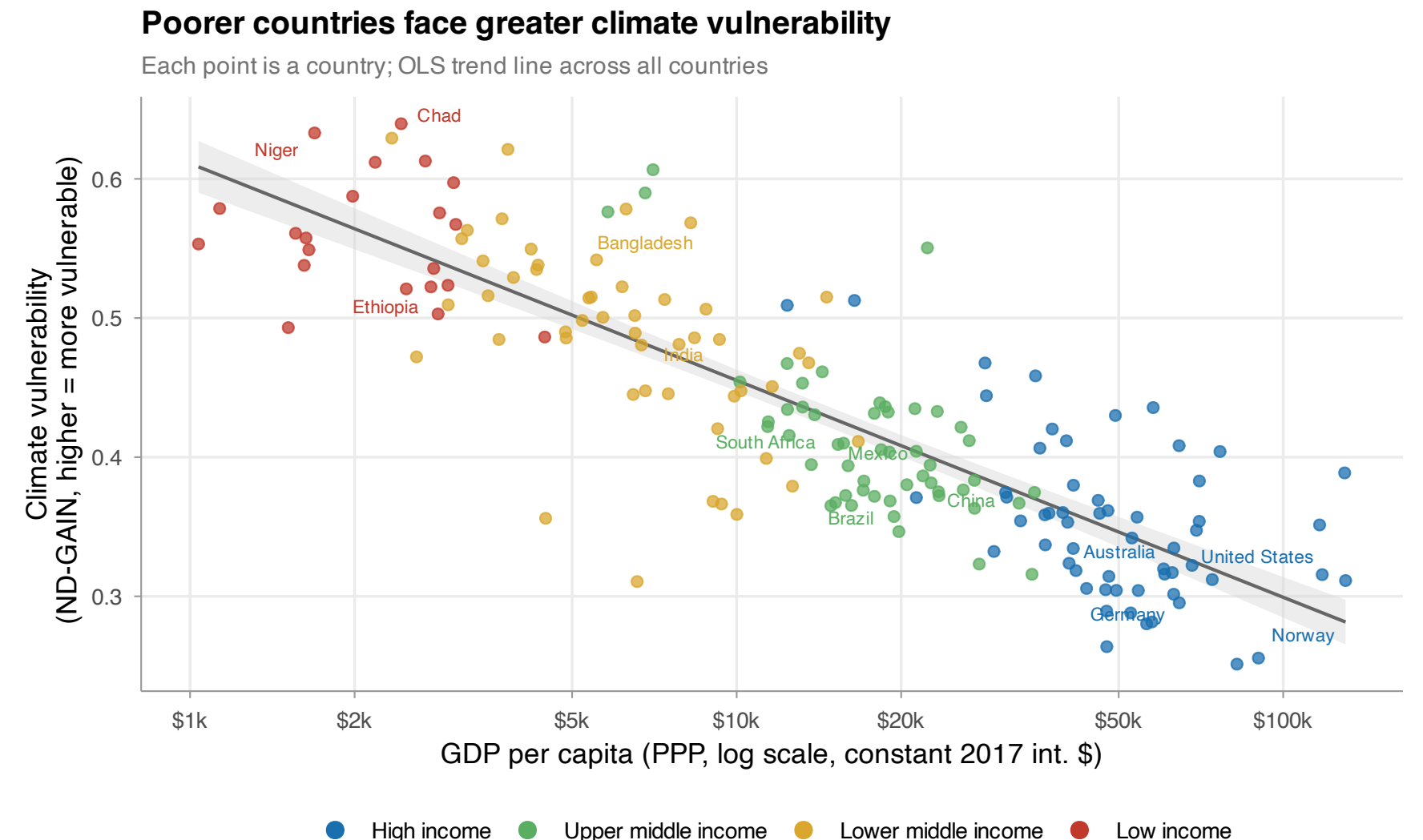
Technological effect

New technologies → different impact per output unit

The channels can work in **opposite directions**; the net result is an empirical and political question – not a fixed law.

Degradation hits developing countries disproportionately

- **Climate change:** floods, droughts, sea-level rise, extreme weather
- **Health costs:** air pollution, water contamination, disease burden
- **Resource depletion:** water scarcity, soil degradation, fishery collapse
- **Economic disruption:** agricultural losses, disaster impact, infrastructure damage



Sources: ND-GAIN Country Index vulnerability score (2023 edition); World Bank WDI, GDP per capita PPP (NY.GDP.MKTP.PP.KD), most recent year 2020–20

Bottom line: poorer people more affected and with fewer resources to adapt

02

The Environmentalism of the Poor

Three types of environmentalism

The Cult of Wilderness

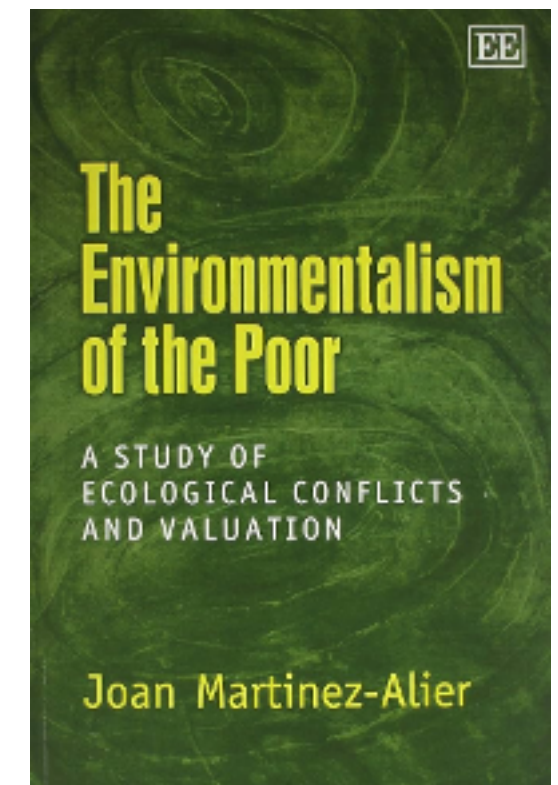
The Gospel of Eco-Efficiency

Environmentalism of the Poor

- For the poor, the environment is a necessity, not luxury
 - Direct source for food, water, livelihood
- Distribution of environmental costs and benefits unequal
 - Ecol. Distribution Conflicts (EDC): local communities vs. corporations
- Losses often defy cost-benefit calculations and compensation
 - Sacred land, health, culture → incommensurable value structures
- Wealthy countries and companies accumulated an ecological debt to the poor



Joan Martinez-Alier



The EoP challenges the key framing

The dominant framing: development vs. Environment

"We must not depress the economy with environmental regulations."

"We must – in the interest of the poor – prioritize social goals, such as poverty reduction, over environmental protection."

The Environmentalism of the Poor

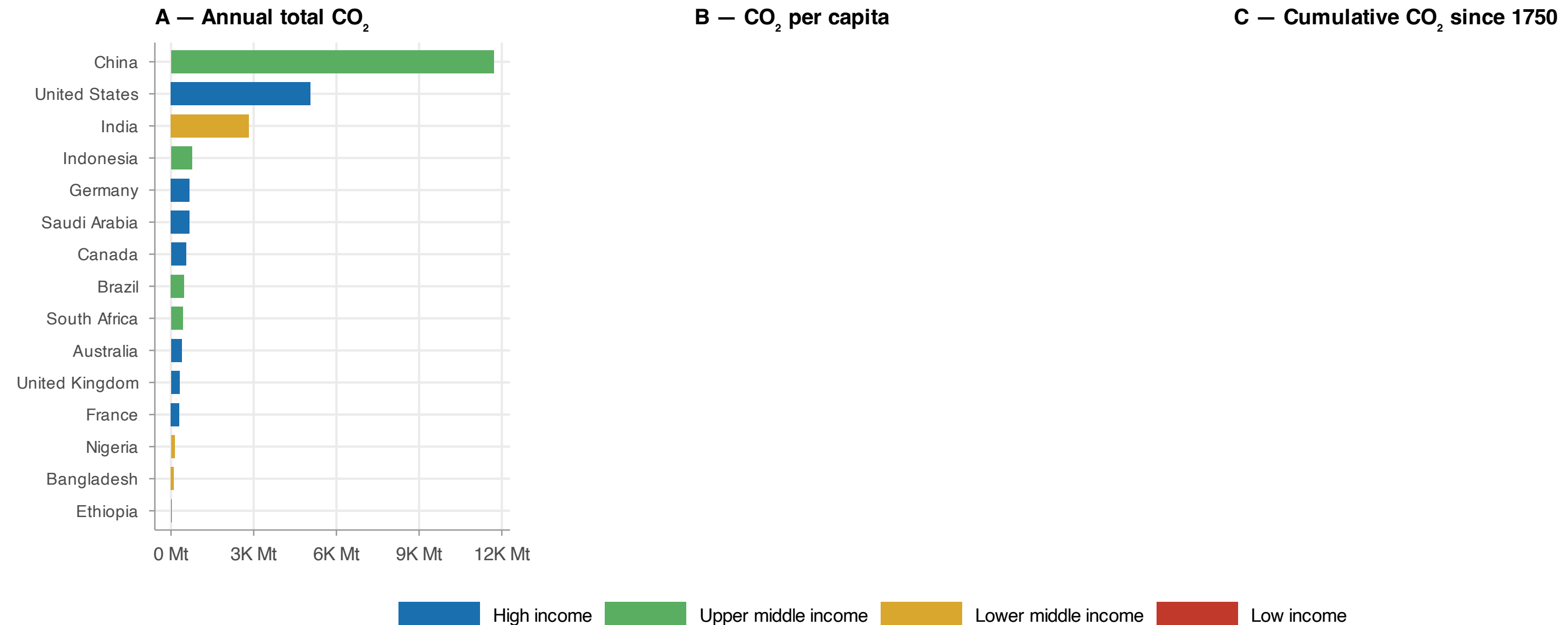
- Degraded environment = degraded livelihood → protection & poverty reduction complementary
- Southern movement often arise from resource conflicts, not post materialism
- Don't ask: "how much growth" → "Whose growth and at whose environmental cost?"

03

What the data does (not) tell us

The value-laddennes of observations again

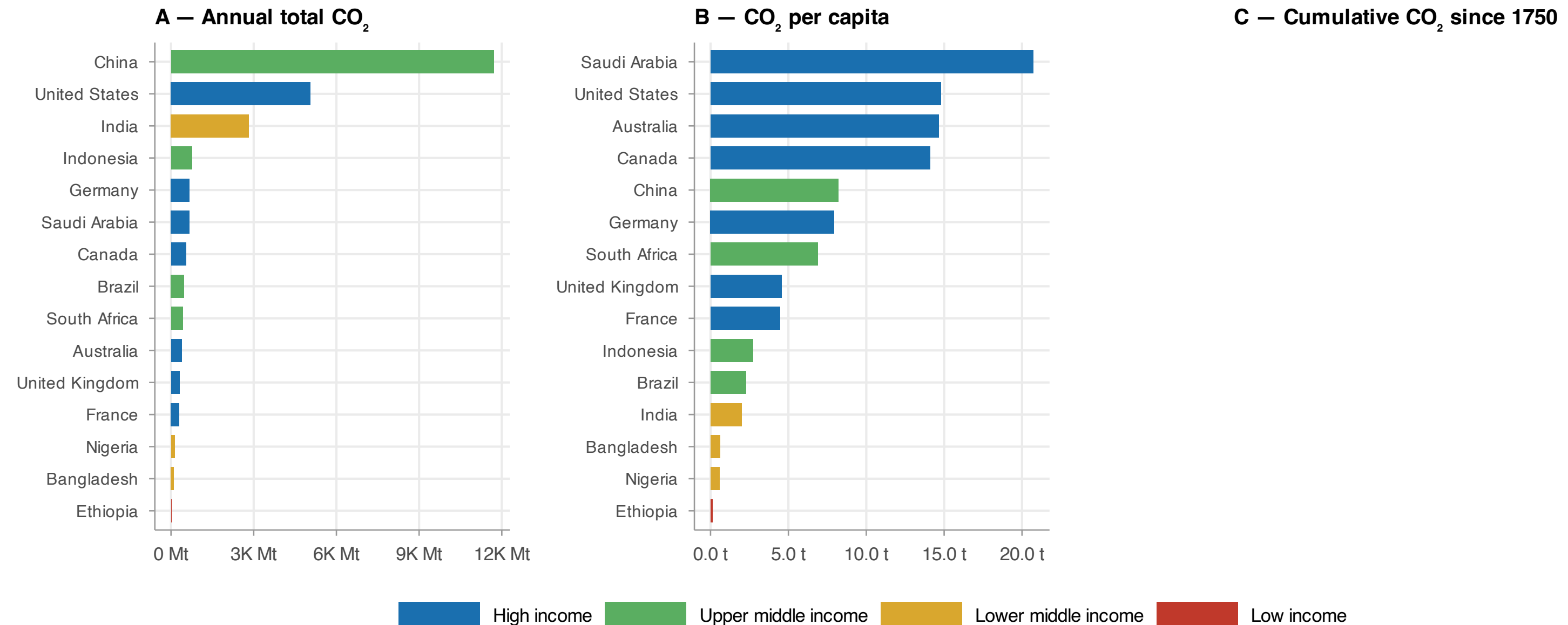
The same emissions, three measures (2022)



Source: Global Carbon Project via Our World in Data (owid-co2-data.csv), year 2022. Income groups: World Bank WDI.

The value-laddennes of observations again

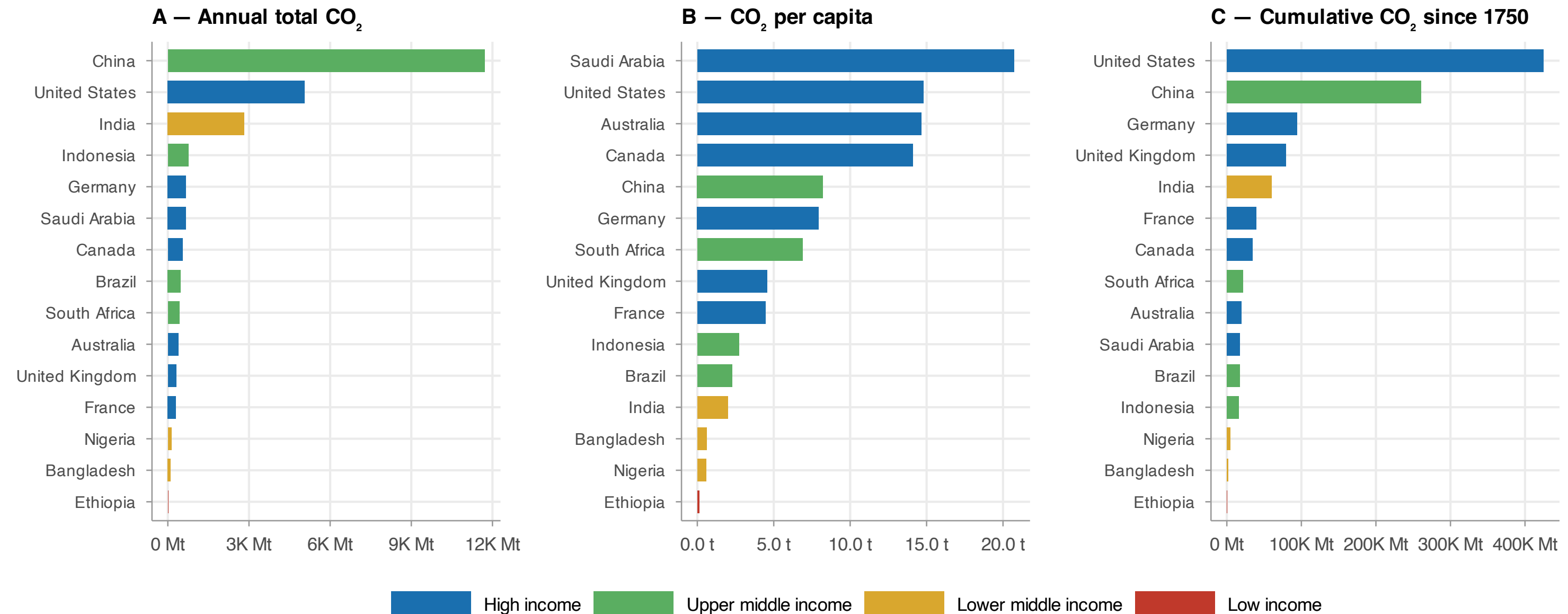
The same emissions, three measures (2022)



Source: Global Carbon Project via Our World in Data (owid-co2-data.csv), year 2022. Income groups: World Bank WDI.

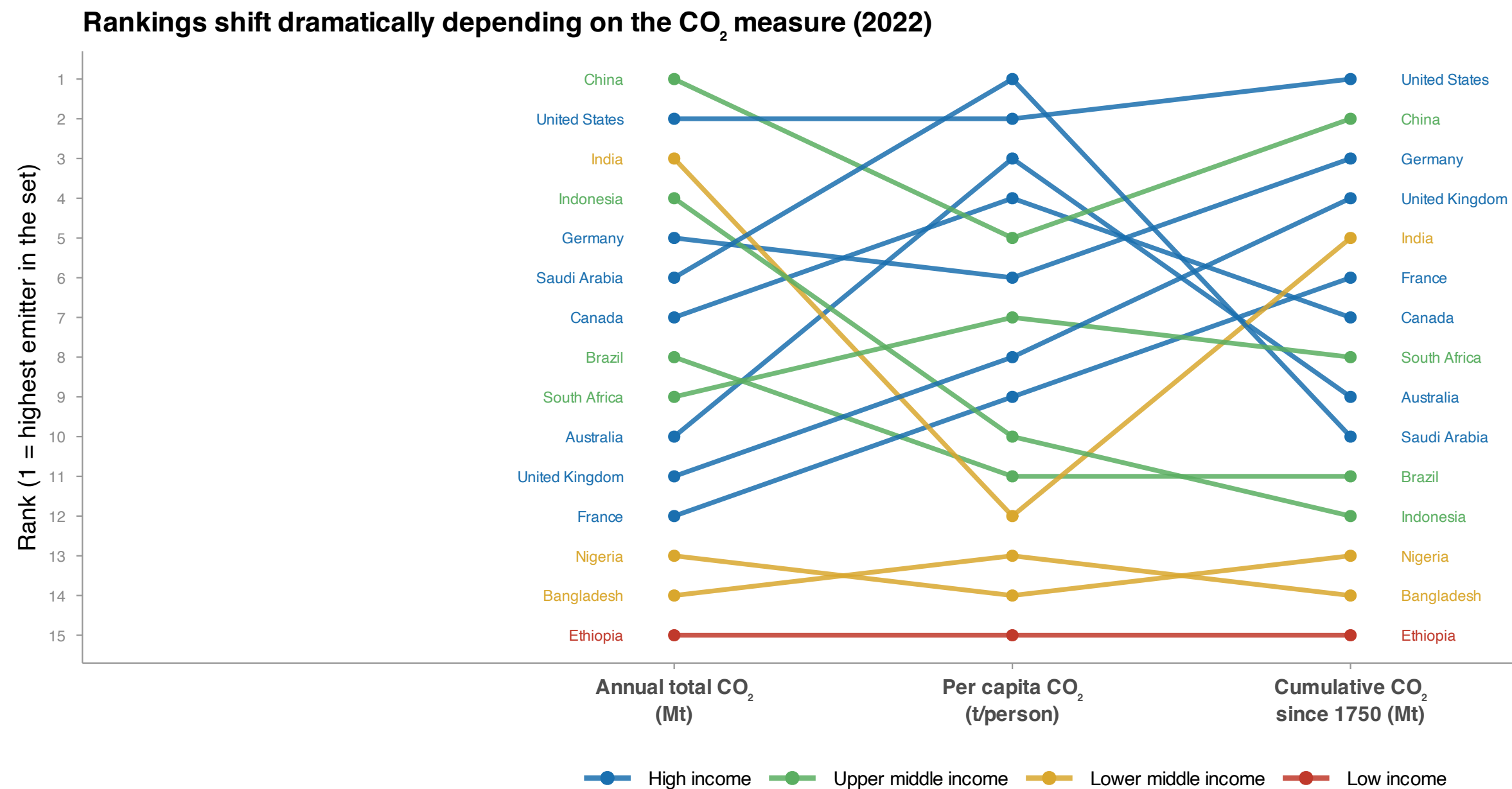
The value-laddennes of observations again

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Source: Global Carbon Project via Our World in Data (owid-co2-data.csv), year 2022. Income groups: World Bank WDI.

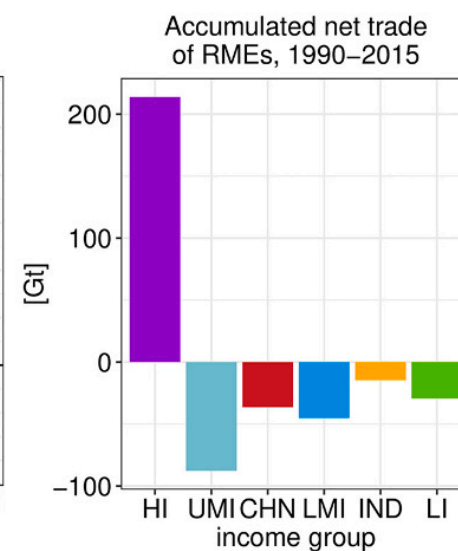
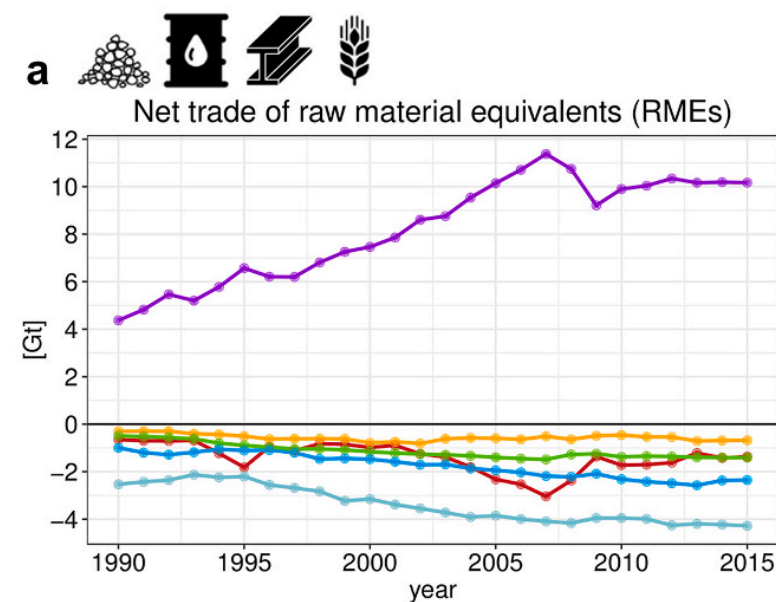
The value-laddennes of observations again



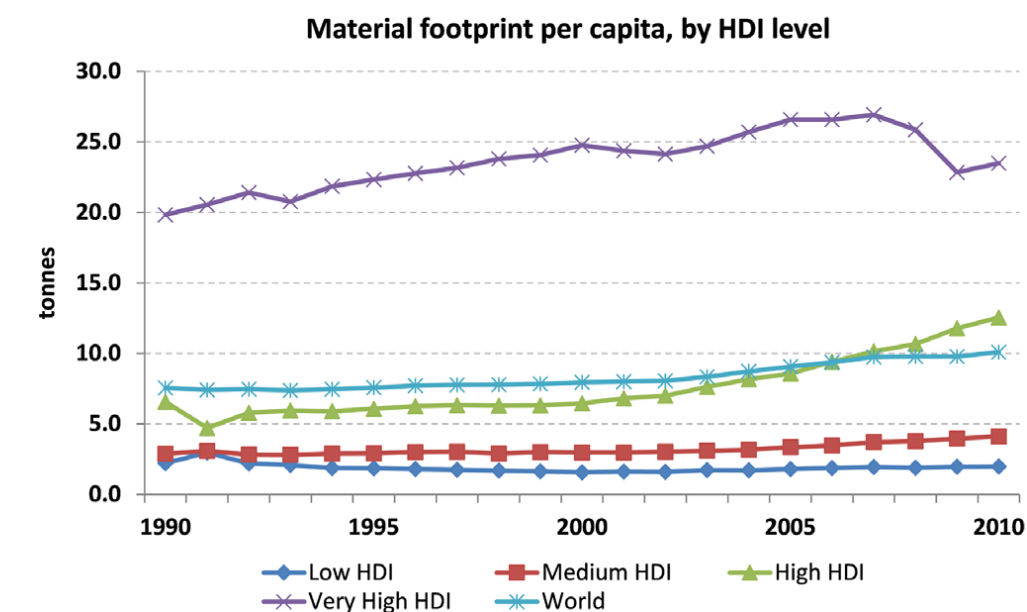
Source: Global Carbon Project via Our World in Data (owid-co2-data.csv), year 2022. Income groups: World Bank WDI.

Resource extraction at the expense of the Global South

- Resource extraction and provision of raw material happen mostly in the Global South
 - Little value-added, high ecological costs



Dorninger et al. (2021)

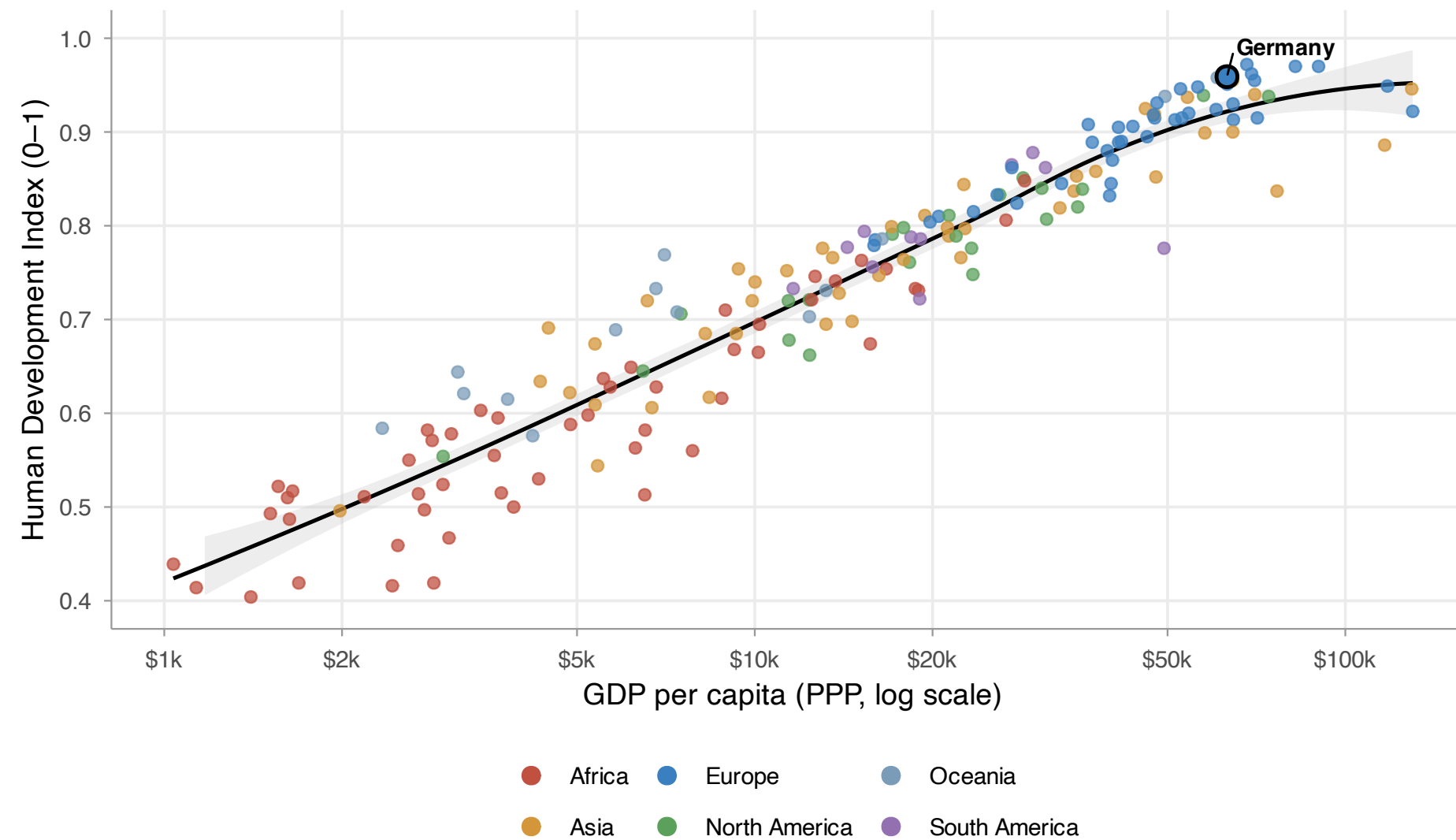


Schandl et al. (2017)

- Compare: dependency and world systems theory on functional differentiation

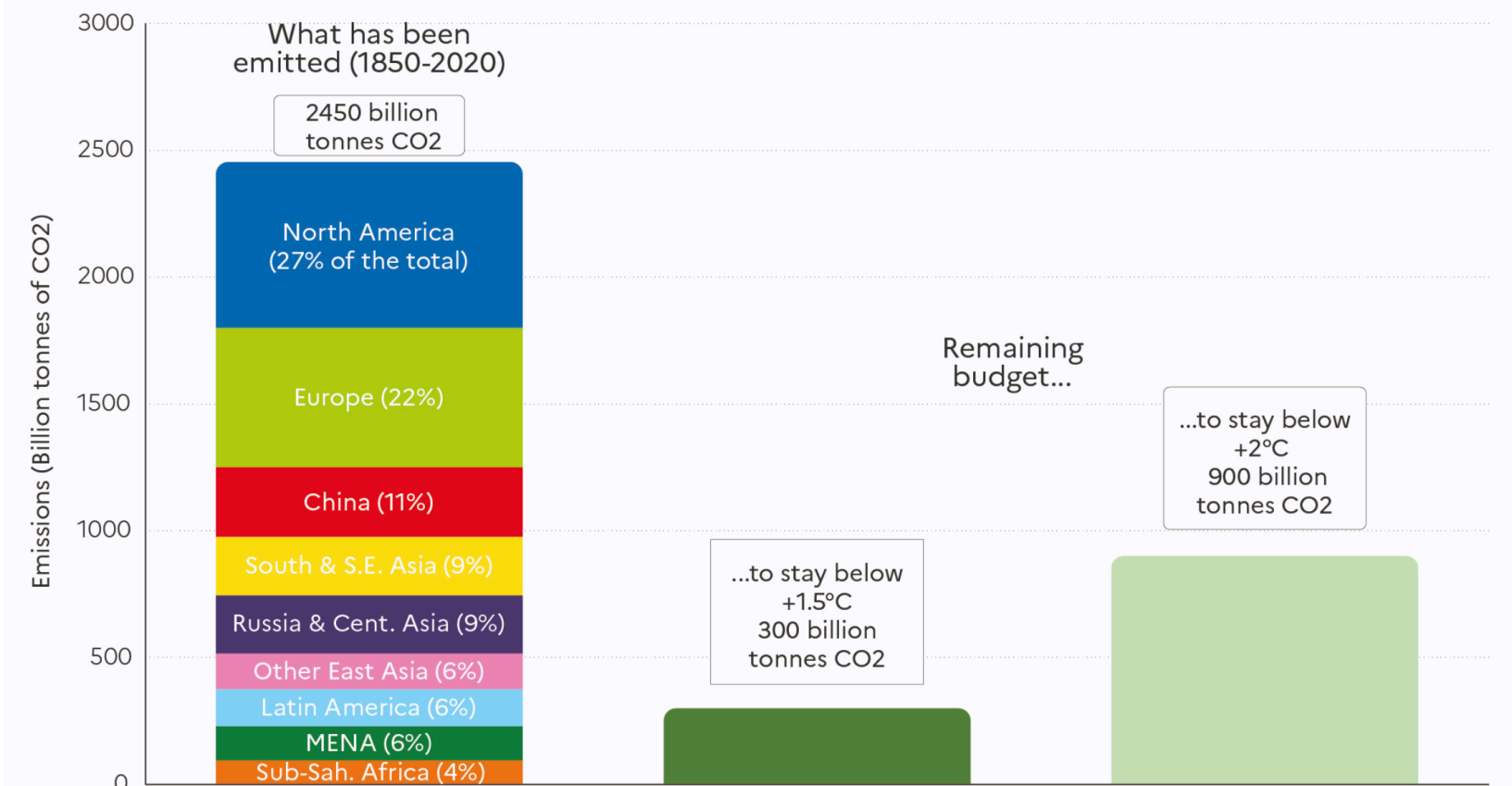
The ambivalence of growth

GDP and human development correlate strongly



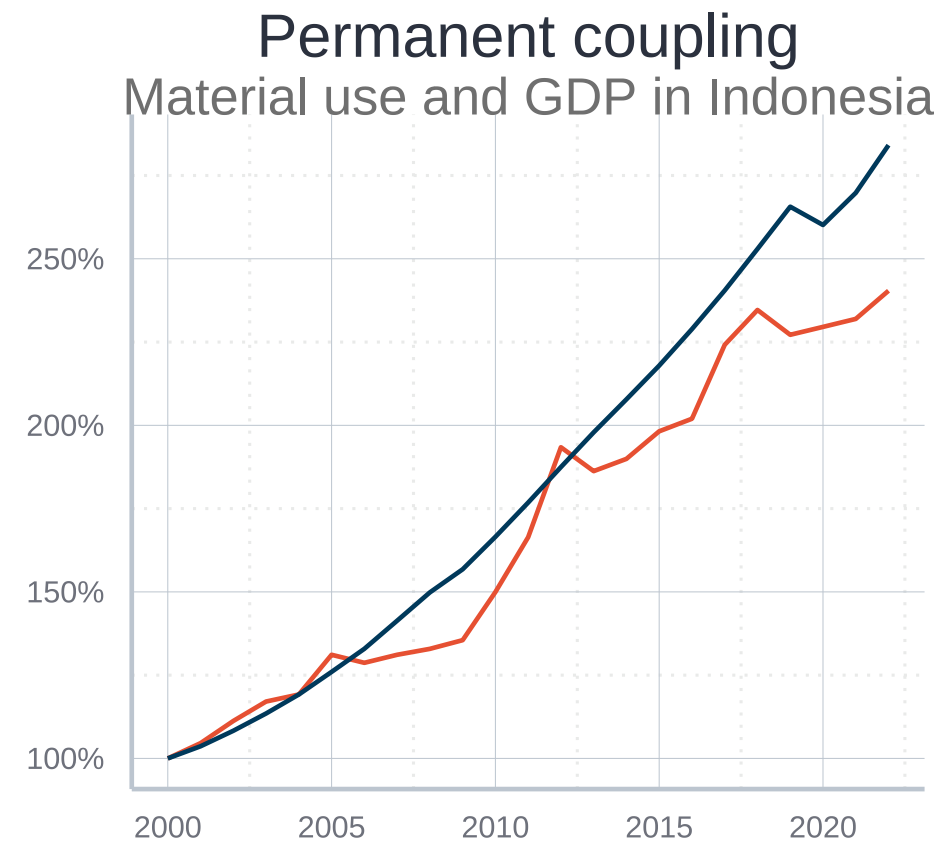
Source: Our World in Data (2023); UNDP Human Development Report.

Figure 6.2 Historical emissions vs. remaining carbon budget



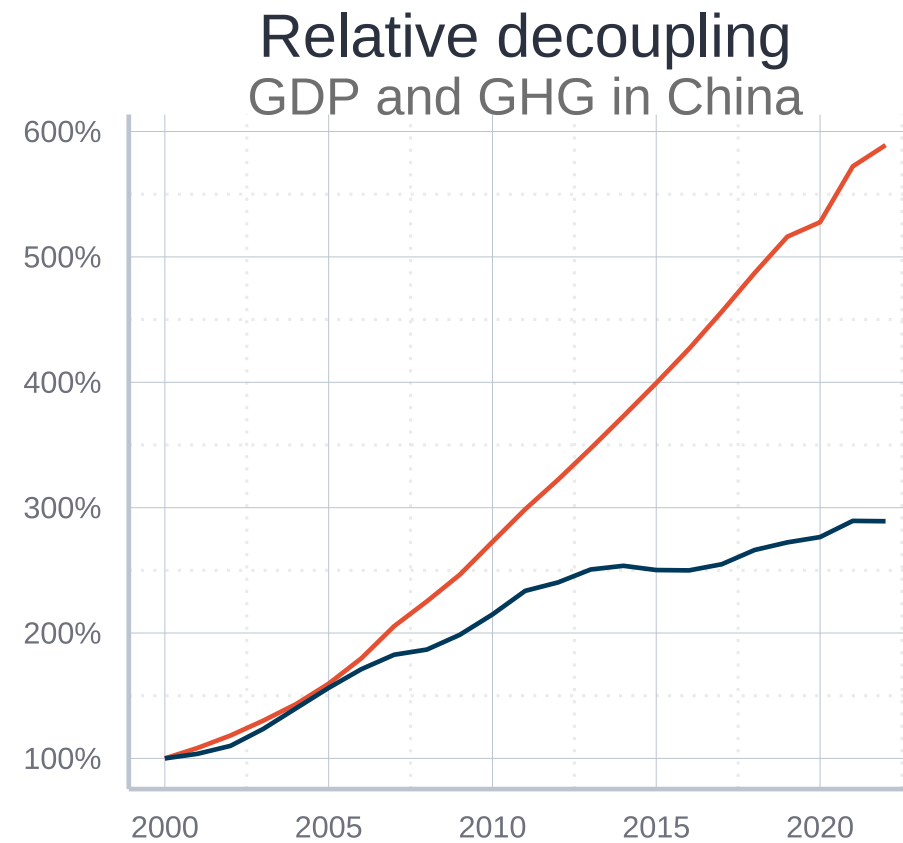
Interpretation: The graph shows historical emissions by region (left bar) and the remaining global carbon budget (center and right bars) to have 83% chances to stay under 1.5°C and 2°C, according to IPCC AR6 (2021). Regional emissions are net of carbon embedded in imports of goods and services from other regions. **Sources and series:** [wir2022.wid.world/methodology](https://www.ourworldindata.org/methodology) and Chancel (2021). Historical data from the PRIMAP-hist dataset.

Growth and the issue of decoupling



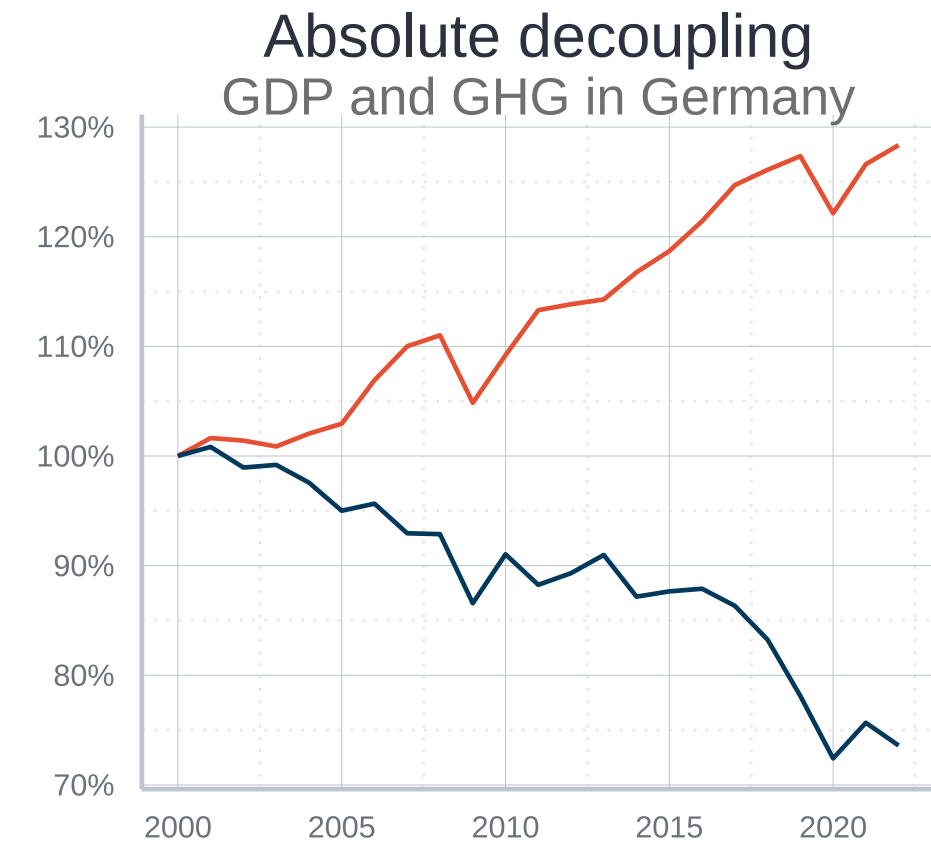
— DMC — GDP

Data: World Bank, Our World in Data.



— GDP — THG

Data: World Bank.



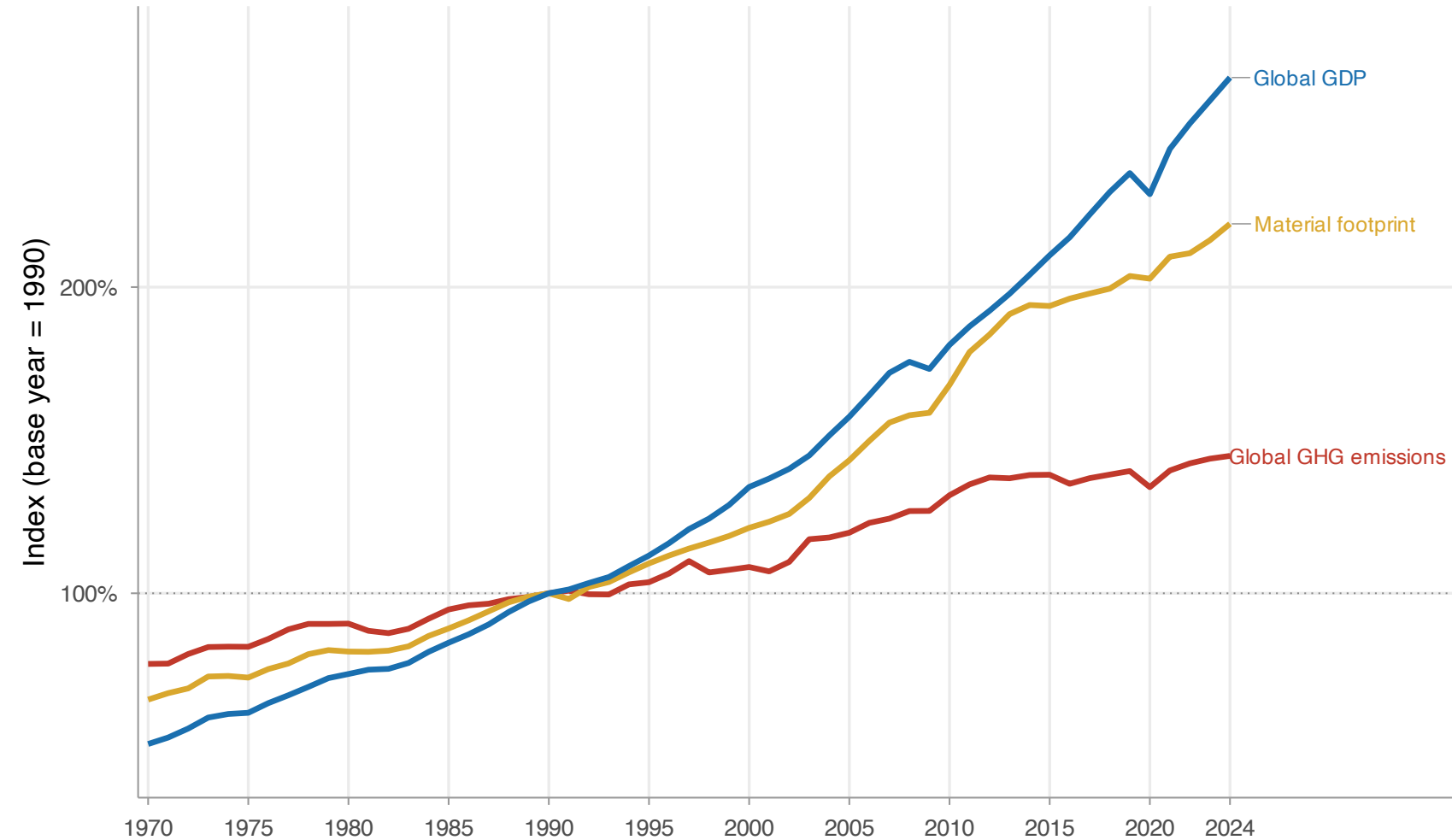
— GDP — THG

Data: World Bank.

DMC - Domestic material consumption
GHG - Greenhouse Gases

Global vs. National decoupling

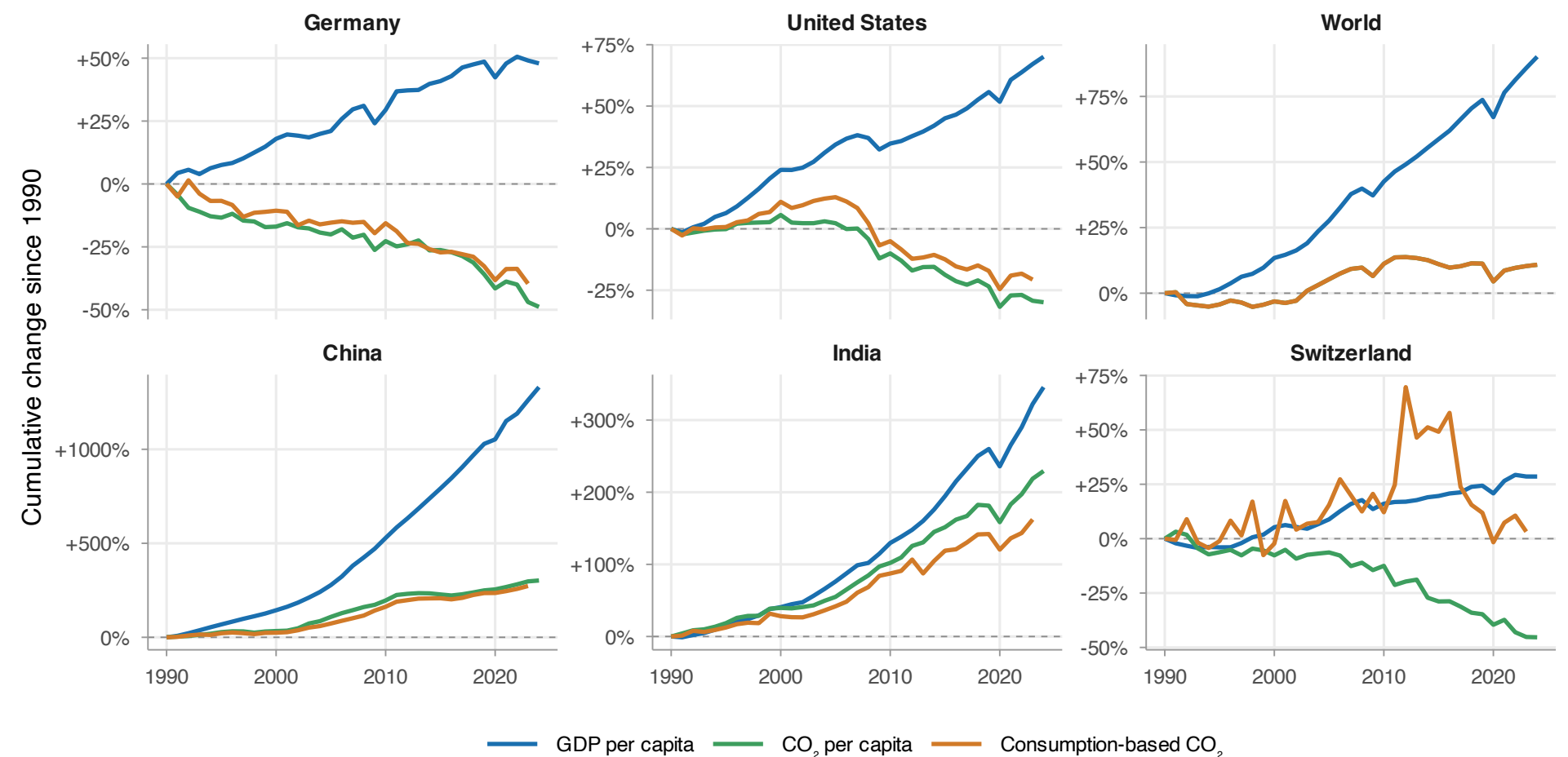
Strong correlation between GDP and resource use



Sources: Global Carbon Project / Jones et al. via Our World in Data (GHG, total_ghg incl. LUC); UNEP IRP Global Material Flows Database (material footprint)

Decoupling of CO₂ emissions from GDP growth since 1990

Cumulative % change since 1990 (base year = 0%)



Source: Our World in Data; Global Carbon Project (2023).

- Important to look at **footprints** to take into account externalization

04

The Doughnut Framework

Vantage point: the planetary boundary framework

- Nine Earth-system processes critical for maintaining **Holocene-like conditions**
- Imply a **"safe operating space"** — within it, civilisation developed over 10,000 years
- **6 of 9 boundaries** already transgressed
 - Holocene → Anthropocene: humanity as geological force

The gap

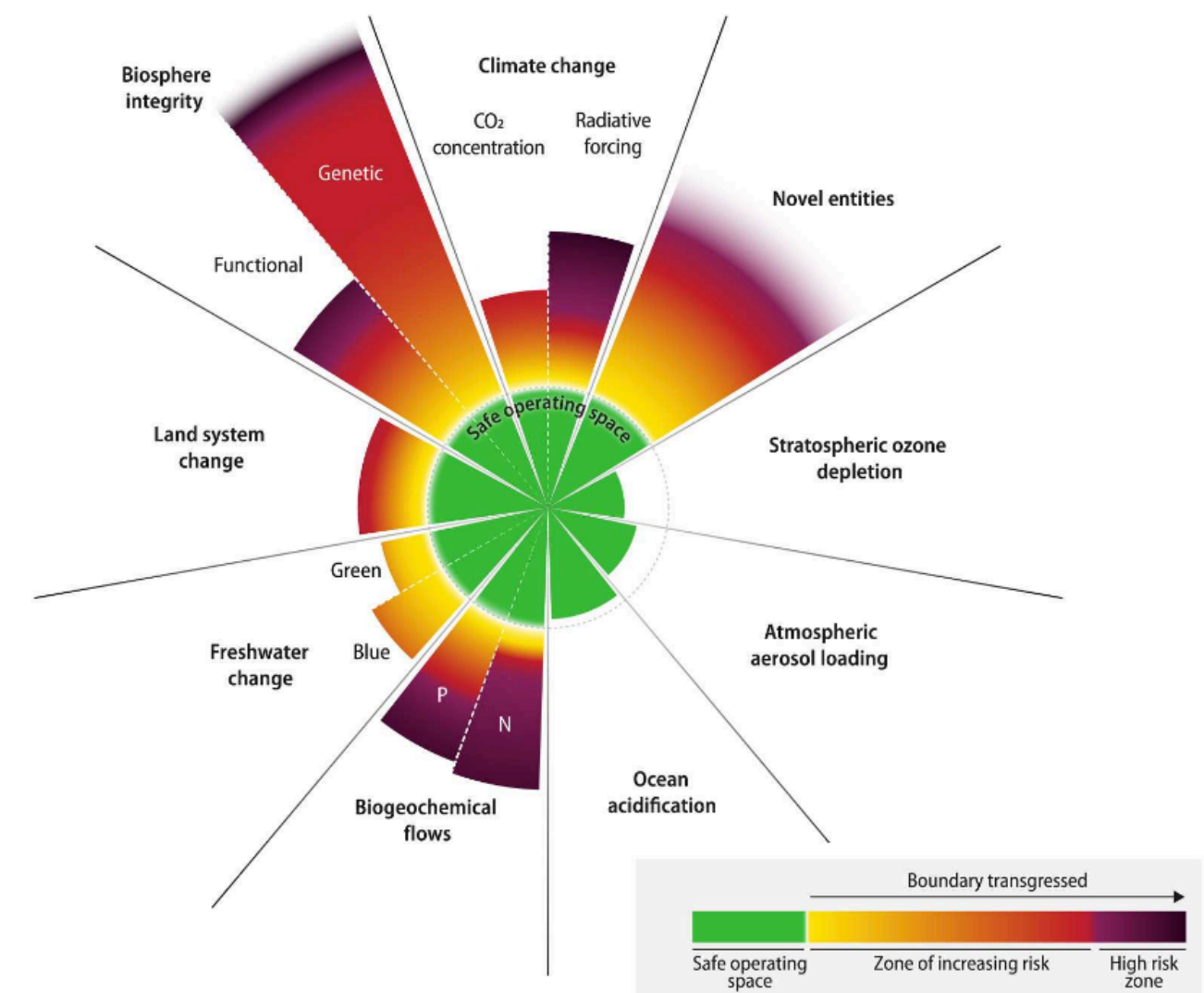
Defines planetary boundaries, remains silent on social justice and human needs.

SCIENCE ADVANCES | RESEARCH ARTICLE

ENVIRONMENTAL STUDIES

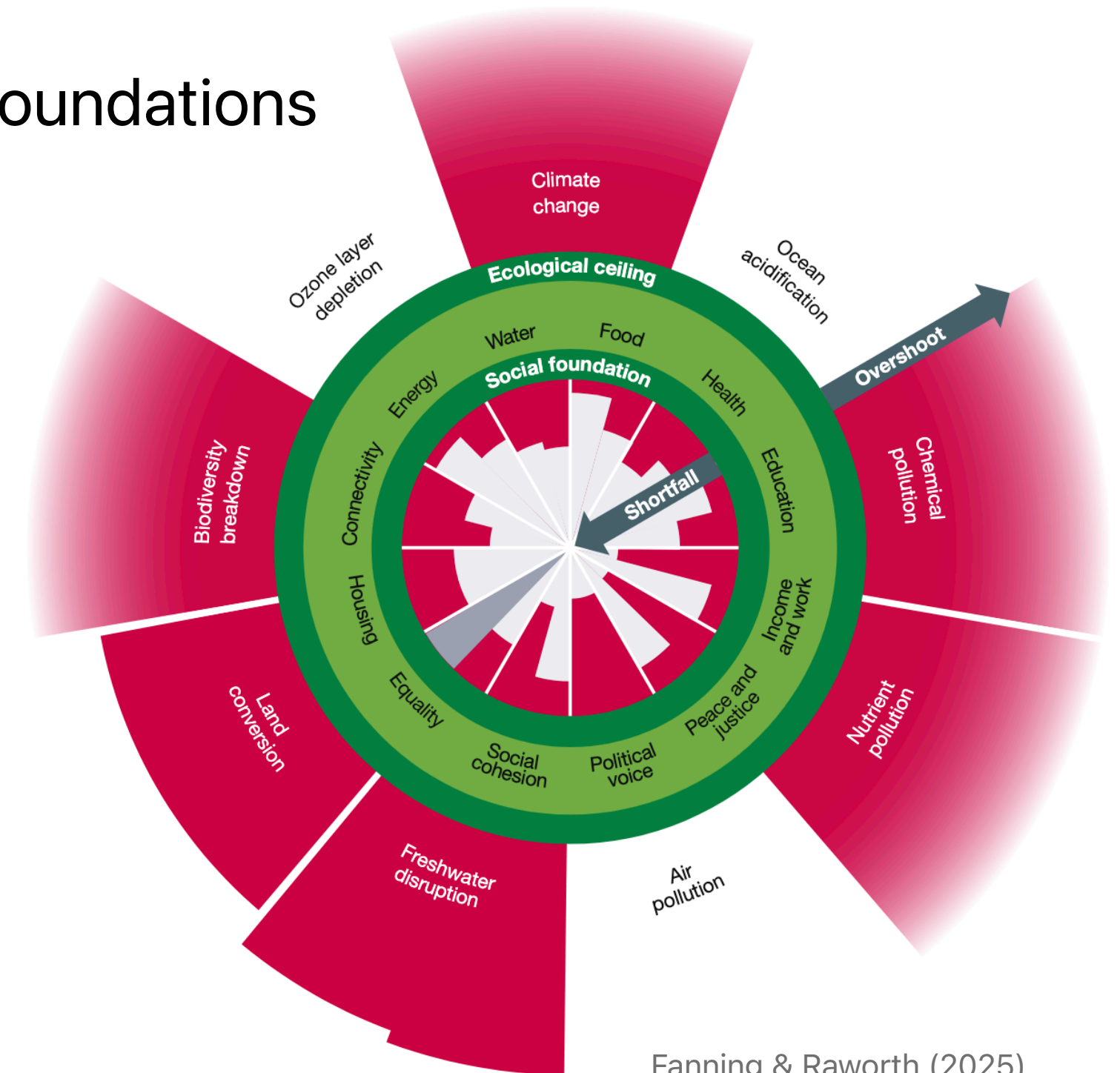
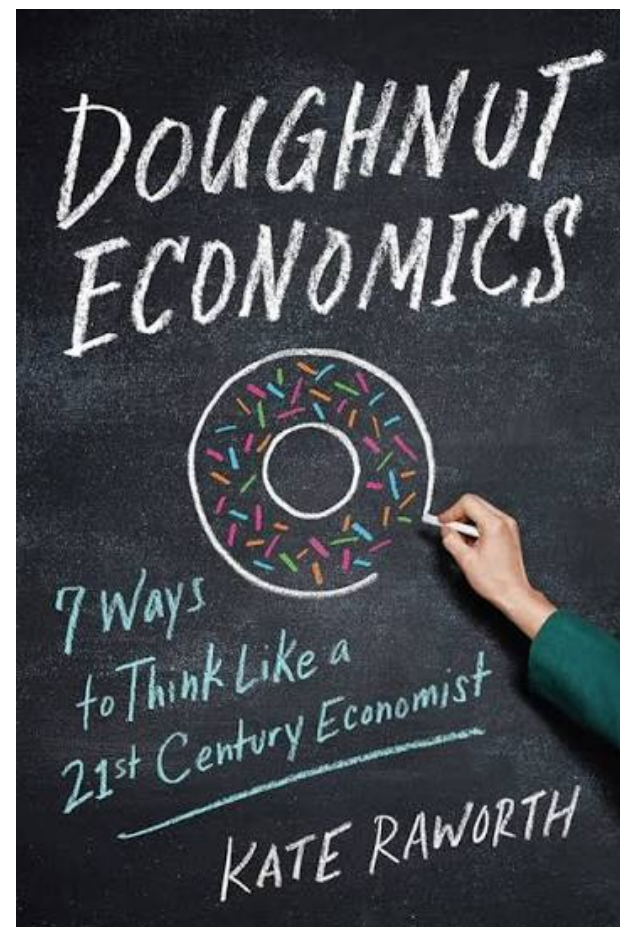
Earth beyond six of nine planetary boundaries

Katherine Richardson^{1*}, Will Steffen^{2†}, Wolfgang Lucht^{3,4}, Jørgen Bendtsen¹, Sarah E. Cornell⁵, Jonathan F. Donges^{3,5}, Markus Drüke³, Ingo Fetzer^{5,6}, Govindasamy Bala⁷, Werner von Bloh³, Georg Feulner³, Stephanie Fiedler³, Dieter Gerten^{3,4}, Tom Gleeson^{9,10}, Matthias Hofmann³, Willem Huiskamp³, Matti Kummu¹¹, Chinchu Mohan^{8,12,13}, David Nogués-Bravo¹, Stefan Petri³, Miina Porkka¹¹, Stefan Rahmstorf^{3,14}, Sibyll Schaphoff³, Kirsten Thonicke³, Arne Tobian^{3,5}, Vili Virkki¹¹, Lan Wang-Erlandsson^{3,5,6}, Lisa Weber³, Johan Rockström^{3,5,15}



Kate Raworth

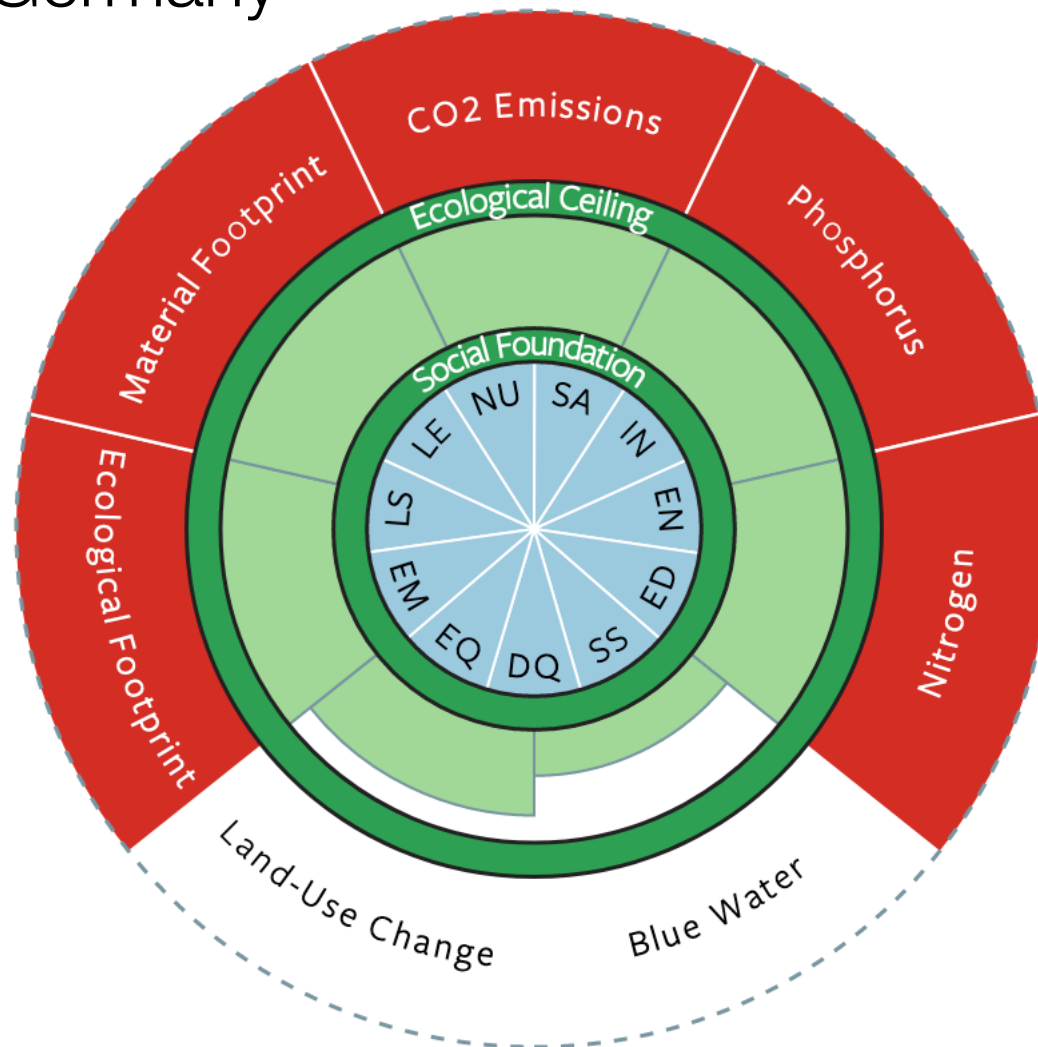
- Goal: complement ecological ceiling with social foundations
- One tool in a broader vision for economics



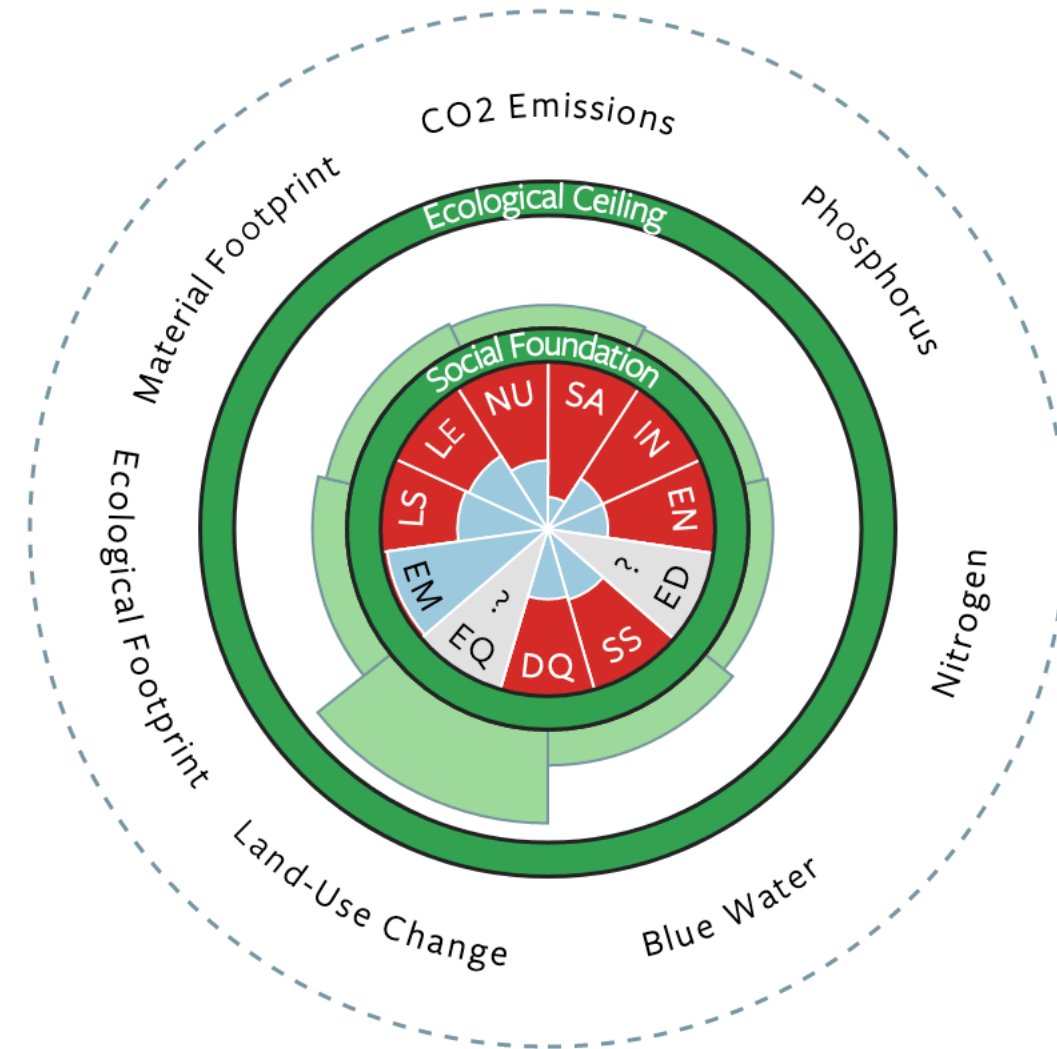
Fanning & Raworth (2025)

Climate inequality through the Donought4 lens

Germany



Haiti

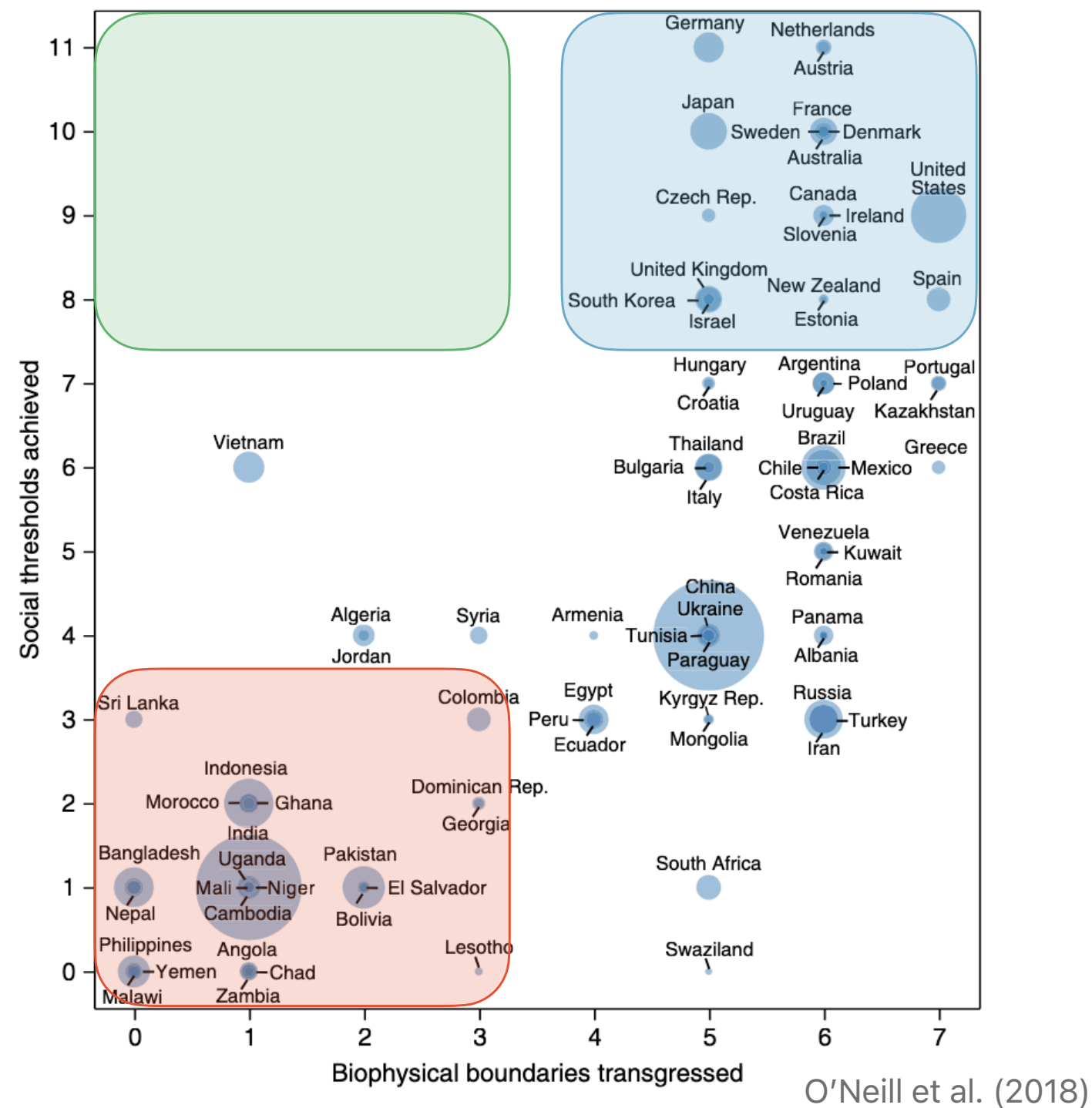


Climate inequality through the Donought lens



Fanning & Raworth (2025)

The dilemma: no country is in the safe and just space



- **High-income** countries:
 - Meet most social thresholds
 - Overshoot ecological ceilings
- **Low-income** countries
 - Comply with ecological ceilings
 - Fall short on the social foundation
- The **safe space** remains empty

Strengths and limitations

Strengths

- Intuitive visualization of social and ecological dimensions
- Conceptual link between earth and social science
- Highlight multi-dimensionality

Weaknesses

- Obscures dependencies
- Focus is on absolute, not cumulative emissions
- Risks of de-politication
- Dynamics less visible
- No immediate tool for predictions

A useful and increasingly institutionalized tool - but not a silver bullet!

05

Growth, Green Growth, and Degrowth

The core debate

- Green growth and degrowth as two extreme positions in a continuous debate

Green growth

- Growth **compatible** with environmental constraints
- Focus on green **technological change**
- Assumption: sufficient **decoupling** feasible

Degrowth

- **Targeted reduction** of throughput (not uniform, ≠recession)
- Focus on **sufficiency**, reparations, redistribution
- Strong **anti-capitalist** tendencies, also a cultural critique

Both accept the need for green transformations, but differ in **ideology** and **scale**.

A case study debate: the role of decoupling

NEW POLITICAL ECONOMY
<https://doi.org/10.1080/13563467.2019.1598964>

 **Routledge**
Taylor & Francis Group

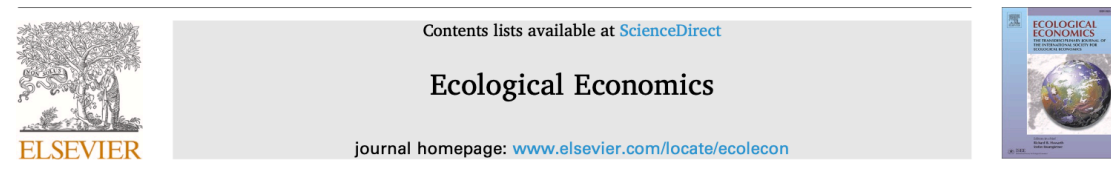


Is Green Growth Possible?

Jason Hickel^a and Giorgos Kallis^b

^aAnthropology, Goldsmiths, University of London, London, UK; ^bICREA and ICTA-UAB, Universitat Autònoma de Barcelona, Barcelona, Spain

Ecological Economics 220 (2024) 108089



COMMENTARY

Confronting the dilemma of growth. A response to [Warlenius \(2023\)](#)

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^a Centre for the understanding of Sustainable Prosperity, University of Surrey, Guildford GU2 7XH, UK

^b Institute of Environmental Science and Technology (ICTA-UAB) and Department of Anthropology, Autonomous University of Barcelona, Barcelona, Spain



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Contents lists available at [ScienceDirect](#)

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journal homepage: www.elsevier.com/locate/ecocon



The limits to degrowth: Economic and climatic consequences of pessimist assumptions on decoupling

Rikard Hjorth Warlenius

Environmental Social Science, School of Global Studies, University of Gothenburg, Sweden

Ecological Economics 232 (2025) 108580



Contents lists available at [ScienceDirect](#)

Ecological Economics

journal homepage: www.elsevier.com/locate/ecocon



Decoupling pessimism and the dilemma of growth. A reply to [Jackson et al. \(2024\)](#)

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^b School of natural science, technology and environmental studies, Södertörn University, 141 89 Huddinge, Sweden

- The debate illustrates several key tensions not open to empirical scrutiny:
 - Is technological or political change faster and more effective?
 - How contingent is the link between economic activity and pollution?

The IPAT Identity

$$I = P \times A \times T$$

Symbol	Meaning	Example
I	Env. Impact	CO ₂ emissions
P	Population	World population
A	Affluence	GDP per capita
T	Technology	Impact per \$ output

- Mathematical identity: true by definition (tautology)
 - Green growth: reduce T
 - Degrowth: reduce A
- **Caution: static equation for dynamic relationships**
 - A and T might be interdependent
 - Relationships might be non-linear

Hickel & Kallis (2020) argue with IPAT

Is Green Growth Possible?

Jason Hickel^a and Giorgos Kallis^b

^aAnthropology, Goldsmiths, University of London, London, UK; ^bICREA and ICTA-UAB, Universitat Autònoma de Barcelona, Barcelona, Spain

Max. Feasible decoupling
4% per year

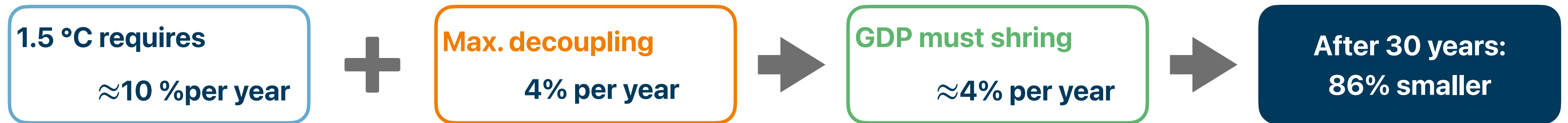
1.5 °C requires
≈10 %per year

Remaining gap → cut A

- **Key evidence:** historical decoupling rates
- **Conclusion:** Global North must degrow, Global South can grow moderately

Warlenius (2023): the implications of HK20

- If decoupling is capped at 4% and targets need ~10% emission reduction, **how much GDP contraction is actually required?**



Assumptions

A and T independent · Contraction is instant, no political economy issues · global incomes converge

Warlenius (2023): the question

Country	GDP pc	After 30 years %
USA	69,287	-98
Germany	51,204	-97
EU	38,234	-96
China	12,556	-89
Brazil	7,519	-82
India	2,277	-39
Kenya	2,007	-31
Least developed	1,177	17

“[a fixed decoupling rate] pushes the observer to either give up on ambitious climate targets, on development and fairness goals — or to assume that developed countries would crash their economies.”

- It is not about giving up climate efforts
- But if decoupling is fixed, he sees little hope → too much degrowth required
- Rather: put all efforts on pushing decoupling
- Implications for degrowth/green growth dichotomy?

The remainder in two quotes

“ [...] the most striking formulation in JHK's comment is that 'Most postgrowth scholars readily accept the need for high levels of decoupling' [...] This statement is fully compatible with the core definition of green growth. [...] This is surprising, as degrowth and green growth are commonly positioned as opposites in discussions on a sustainable future.”

“ Our differences revolve around the question of capitalism: can it be expected to stop growing, and is it likely to disappear within short? If the answer to both is no, strong decoupling is our best chance to avoid runaway climate change.

06

Discussion

Joint discussion

Based on what you have heard today: is economic growth primarily the **problem**, the **solution**, or **neither** — for the development–environment nexus?

- 1 What is the most important argument for you? Which fact would change your mind?
- 2 Does your answer depend on the country considered?
- 3 What is your most pressing open question?

Think 2 minutes on your own

Talk 5 minutes with your neighbors

Short wrap-up in the plenary

07

Synthesis and outlook

Summary & outlook

- **Development and environment nexus:** structural but contingent relationship
- **Environmentalism of the Poor:** environmental protection and poverty reduction are complementary, not substitutive
- The **data** cannot resolve political debates, the past does not determine the future
 - Absolute decoupling feasible for emissions, but fragile for materials
- The **doughnut** as an appealing but potentially depoliticising visualization
- The **degrowth debate** cannot be settled by data alone
 - Fundamental question: malleability of decoupling processes and desirability of capitalism